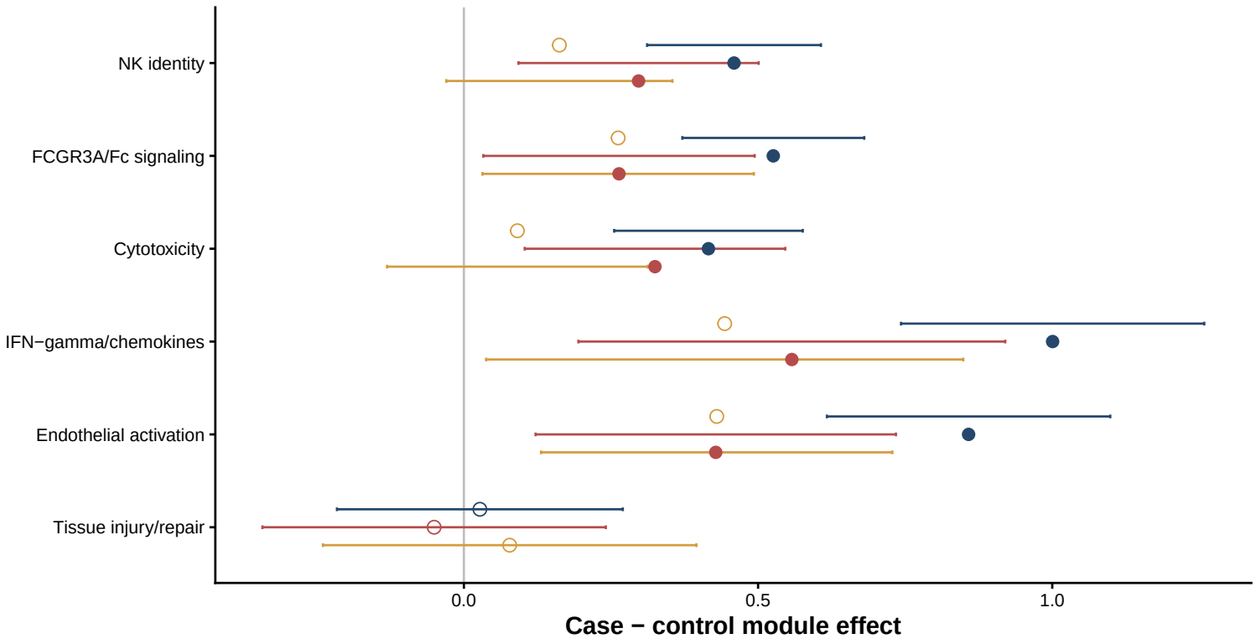
MOSAIC−Tx: qualification of a treatment−responsive rejection program

Effect estimates answer prespecified biological questions; descriptive PCA is excluded from the primary result

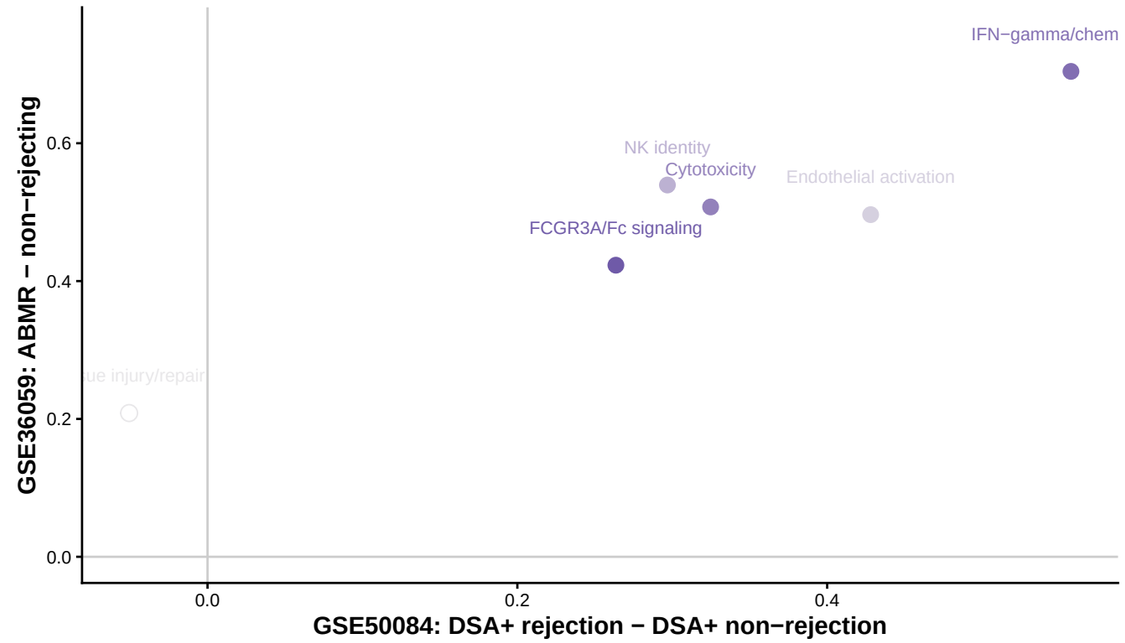
A DSA exposure is separated from damaging rejection

GSE50084, 61 primary biopsies; standardized module effects with approximate 95% CI



B Replication is challenged for rejection specificity

Color: ABMR − TCMR; purple means the program is stronger in TCMR



Estimand DSA exposure without rejection · Rejection among DSA+ grafts · DSA+ rejection vs DSA− control

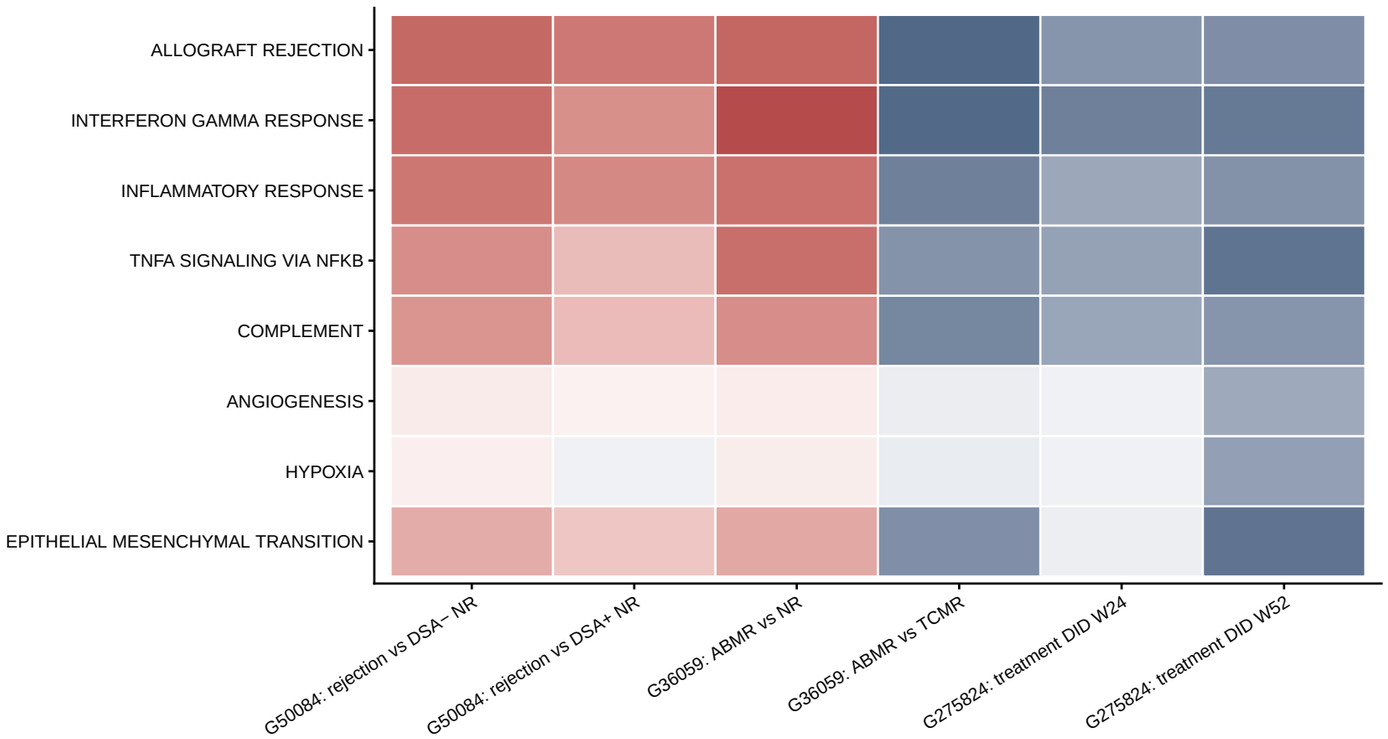
Evidence FDR < 0.05 · FDR >= (

ABMR − TCMR −0.8 −0.6 −0.4 −0.2

Replicated FALSE · TRUE

C Pathway direction across discovery, specificity and treatment

Red: enriched in first group; blue: enriched in control/placebo direction; opacity denotes BH FDR

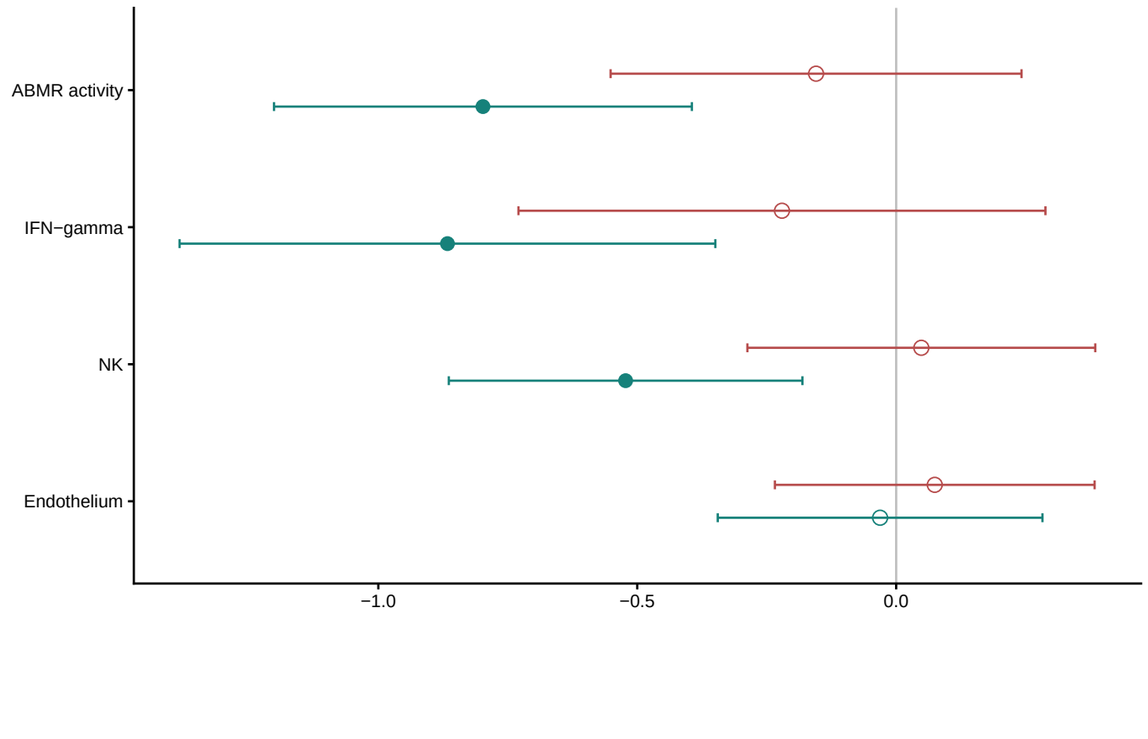


FDR < 0.05 FALSE · TRUE

Hallmark NES −2 0 2 4

D Randomized treatment effects are transient and program−dependent

Felzartamab−minus−placebo difference−in−differences; negative values indicate relative reduction



Time Week 24 · Week 52

Evidence FDR < 0.05 · FDR >= 0.05

Associative bulk transcriptomics cannot prove antibody engagement, cell of origin, target causality, or drug efficacy.